جامعة الفيصل
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# Introduction to Artificial Intelligence (SE 444)

## Lecture 10: Reasoning Under Uncertainty

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# Two Visions of Intelligence

## Certainty through Logic

- Deduction from axioms
- Binary truth values
- Mathematical proof
- Complete knowledge assumption

## Confidence through Probability

- Induction from observations
- Degrees of belief [0,1]
- Statistical inference
- Partial knowledge handling

Which vision better captures real-world intelligence?



# Knowledge vs. Faith Analogy

## Human Knowledge

**Absolute truth** through reason

## Logic-Based AI

**Certainty** through symbolic reasoning, axioms, proof

## Rational Faith

**Trust** informed by evidence

## Probabilistic AI

Acts rationally under uncertainty

# Two Schools of AI

## Logicist/Symbolic School

*"Intelligence is reasoning with certainty"*

**Philosophers:** Aristotle, Descartes, Boole, Russell & Whitehead

**AI Pioneers:** McCarthy, Minsky, Simon & Newell

**Methods:** First-order logic, expert systems, theorem proving

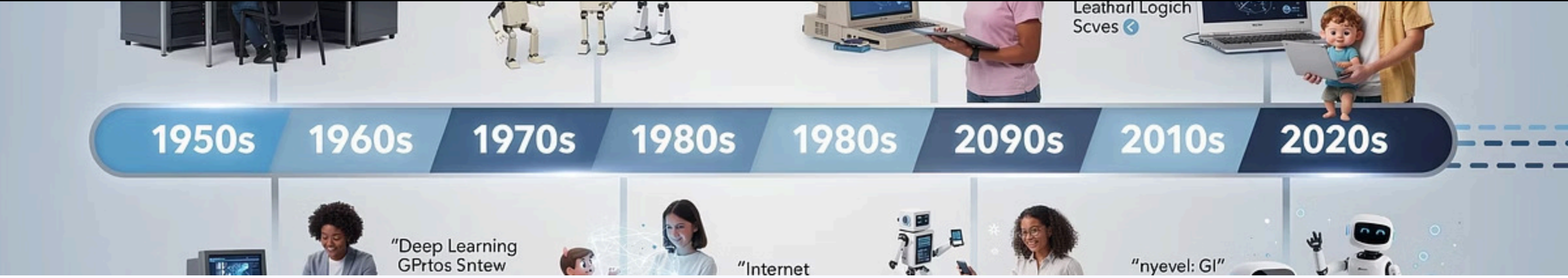
## Probabilistic School

*"Intelligence is reasoning under uncertainty"*

**Philosophers:** Hume, Bayes, Laplace, de Finetti

**AI Pioneers:** Pearl, Russell, Thrun, Jordan

**Methods:** Bayesian networks, machine learning, probabilistic robotics




# Historical Evolution of AI

### 1950s-60s: Logic Era

Logic Theorist, GPS, symbolic manipulation

1

### 1970s-80s: Expert Systems

MYCIN, DENDRAL–struggled with uncertainty

2

### 1980s-90s: Probabilistic Revolution

Bayesian Networks (Pearl, 1988)–paradigm shift

3

### 2000s+: ML Era

Deep learning, fundamentally probabilistic AI

4

# Modern Unified View

> *"Symbolic AI pursues truth. Probabilistic AI pursues confidence. Modern AI seeks to combine both."*

## When Logic Excels

- Complete information
- Deterministic environments
- Formal verification needed
- Guaranteed correctness required

*Example: Software verification*

## When Probability Excels

- Incomplete information
- Noisy sensors
- Stochastic environments
- Learning from data

Example: Robotics, vision, NLP

Logic assumes a world it does not inhabit



# Logic's Requirements vs. Reality

1

## Logic Requires

**Complete** knowledge, **perfect** information, **deterministic** rules, **exact** truth values

2

## Reality Provides

**Partial** knowledge, **uncertain** observations, **stochastic** outcomes, **continuous** degrees

The real world is inherently uncertain–sensors are noisy, observations incomplete, outcomes probabilistic

# The Qualification Problem

**Definition:** The impossibility of listing all preconditions for an action to succeed in the real world

**Logical Rule:**

$$\text{Outside} \land \text{Raining} \rightarrow \text{Wet}$$

**But what if...** umbrella? raincoat? covered walkway? rain stops? too light?

Exhaustively listing all conditions is impossible–logic breaks down

# Five Types of Uncertainty

01

## Laziness

Too many conditions to list explicitly

02

## Theoretical Ignorance

No complete theory exists (e.g., disease mechanisms)

03

## Practical Ignorance

Information not accessible (future events)

04

## Sensor Noise

Observations are imperfect and noisy

05

## Adversarial

Opponents' actions unpredictable

# The Paradigm Shift

From **knowing** to believing



# Old vs. New Paradigm

## Old Paradigm: Logic

Represent knowledge as facts and rules. Deduce new facts with **certainty**. Works when: complete info, deterministic world

## New Paradigm: Probability

Represent knowledge as **degrees of belief**. Update beliefs as evidence accumulates. Works in reality: incomplete info, uncertain world

Rational agents act to maximize expected utility under uncertainty




# Subjective vs. Objective Probability

## Frequentist (Objective)

Probability = long-run frequency

$$P(\text{Heads}) = \lim_{n \to \infty} \frac{\text{count}(\text{Heads})}{n}$$

**Problem:** What about one-time events?

## Bayesian (Subjective)

Probability = degree of belief based on current knowledge

**Advantages:** Applies to any proposition, updates with evidence, rational decision framework

# Probability Fundamentals

## Sample Space

Set of all possible outcomes

$$\Omega$$

Example: $\Omega = \{H, T\}$

## Event

Subset of sample space

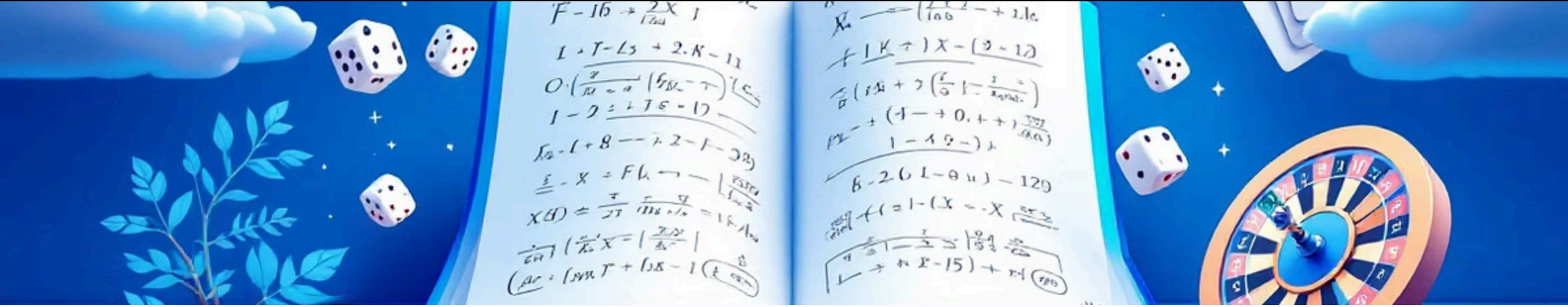
$$A \subseteq \Omega$$

Example: $A = \{H\}$

## Probability Function

Maps events to [0,1]

$$P : A \to [0, 1]$$



# Kolmogorov's Three Axioms

### Axiom 1: Non-negativity

$$P(A) \geq 0$$

Probabilities cannot be negative

### Axiom 2: Normalization

$$P(\Omega) = 1$$

Something must happen

### Axiom 3: Additivity

$$P(A \cup B) = P(A) + P(B)$$

If $A \cap B = \emptyset$

All of probability theory follows from these three axioms!

# Core Probability Concepts

1

## Joint Probability

$$P(A, B) = P(A \cap B)$$

Probability that both A and B occur

2

## Conditional Probability

$$P(A|B) = \frac{P(A, B)}{P(B)}$$

Probability of A given that B occurred

3

## Marginal Probability

$$P(A) = \sum_{b} P(A, B = b)$$

Probability of A regardless of B

# The Product Rule (Chain Rule)

Derivation from conditional probability:

$$P(A|B) = \frac{P(A,B)}{P(B)}$$

Rearranging:

$$P(A,B) = P(A|B) \cdot P(B) = P(B|A) \cdot P(A)$$

**General Chain Rule:**

$$P(X_1, X_2, \ldots, X_n) = P(X_1) \cdot P(X_2|X_1) \cdot P(X_3|X_1, X_2) \cdots P(X_n|X_1, \ldots, X_{n-1})$$



# Medical Diagnosis Example

**Scenario:** Patient has disease D and symptom S

| 1% | 90% | 20% |
|---|---|---|
| P(D) | P(S\|D) | P(S\|¬D) |
| Disease prevalence | Sick people with symptom | Healthy people with symptom |

**Question:** If someone has the symptom, what's $P(D|S)$?




# Bayes' Theorem

The Most Important Formula in AI

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

# Anatomy of Bayes' Rule

1 2 3

## Prior

$P(A)$

What we believed before

## Likelihood

$P(B|A)$

How likely is evidence given hypothesis

## Posterior

$P(A|B)$

Updated belief after evidence

**Intuitive Form:**

$$\text{Posterior} = \frac{\text{Likelihood} \times \text{Prior}}{\text{Evidence}}$$

# Solving Medical Example with Bayes

### Step 1: Compute P(S) using Law of Total Probability

$$P(S) = P(S|D) \cdot P(D) + P(S|\neg D) \cdot P(\neg D)$$

$$P(S) = 0.9 \times 0.01 + 0.2 \times 0.99 = 0.207$$

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### Step 2: Apply Bayes' Rule

$$P(D|S) = \frac{P(S|D) \cdot P(D)}{P(S)} = \frac{0.9 \times 0.01}{0.207}$$

$$P(D|S) = 0.0435 \ (4.35\%)$$



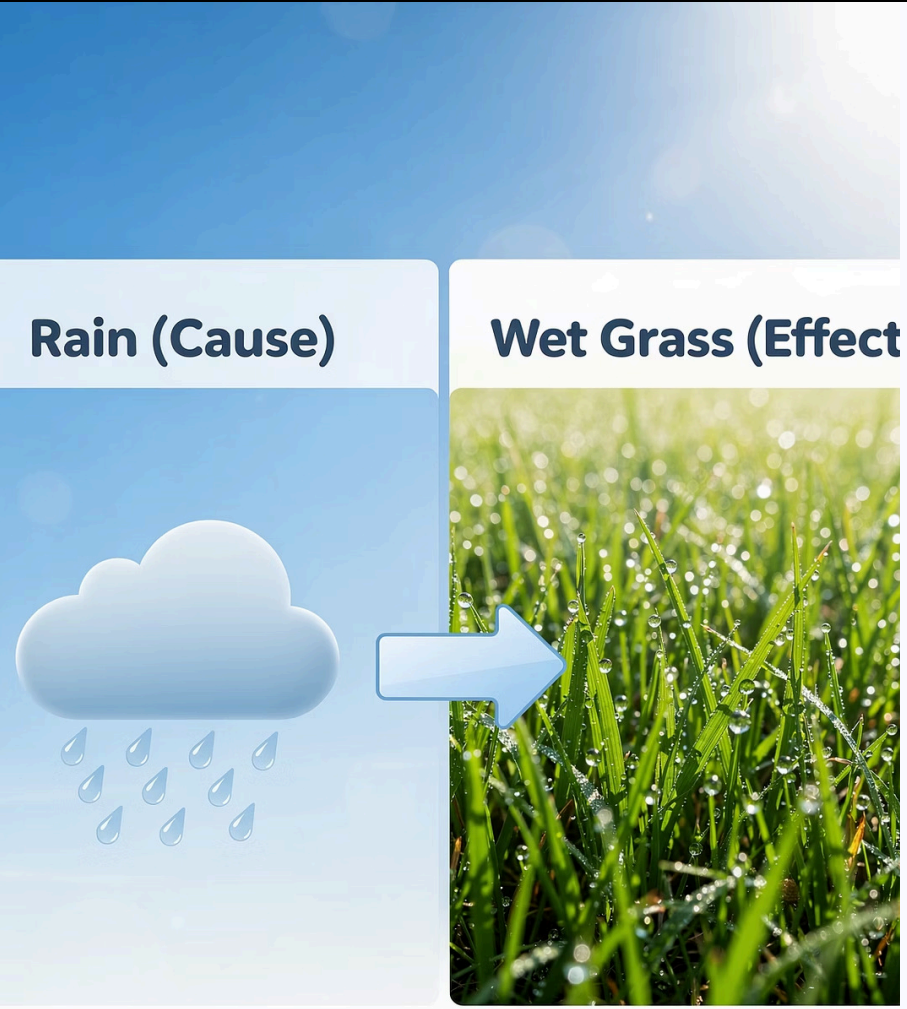
# The Base Rate Fallacy

Even with a 90% accurate test, testing positive gives only 4.35% chance of disease!

**Why?**

- Disease is **rare** (base rate = 1%)
- Test has **false positives** (20% of healthy people)
- Most positive results come from the larger healthy population

> Base rates matter! People often ignore them and focus only on test accuracy. Bayes' Rule forces us to account for **prior probabilities**.




# Diagnostic vs. Causal Reasoning

## Causal Reasoning

From cause to effect

$$P(\text{Effect}|\text{Cause})$$

*"If it rains, probability grass is wet?"*

Usually **easy to estimate** (we understand causation)

## Diagnostic Reasoning

From effect to cause

$$P(\text{Cause}|\text{Effect})$$

*"If grass is wet, probability it rained?"*

Usually **hard to estimate** (reverse inference difficult)

Bayes' Rule converts easy causal probabilities to hard diagnostic probabilities




# Independence

**Definition:** Two events A and B are **independent** if:

$$P(A, B) = P(A) \cdot P(B)$$

Equivalently: $P(A|B) = P(A)$ (knowing B doesn't change belief about A)

**Conditional Independence:** A and B are conditionally independent given C if $P(A, B|C) = P(A|C) \cdot P(B|C)$

# Common Probability Distributions

1

## Bernoulli

Single binary outcome

$$P(X = 1) = p$$

Example: Single coin flip

2

## Binomial

Number of successes in n trials

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Example: Heads in 10 flips

3

## Normal (Gaussian)

Continuous, bell curve

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Example: Heights, errors

# Central Limit Theorem

One of the most important theorems in statistics

The sum or average of many independent random variables tends toward a normal distribution, **regardless of the original distributions**

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1)$$

**Why it matters:**

- Explains why normal distributions appear everywhere
- Foundation for statistical inference
- Justifies many ML assumptions

# Law of Large Numbers

**Strong Law of Large Numbers:** As $n \to \infty$, sample average converges to expected value

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{a.s.} \mathbb{E}[X] = \mu$$

**Intuition:**

- Flip coin many times: proportion of heads → 0.5
- Roll die many times: average value → 3.5

This theorem connects probability to reality—justifies using frequencies to estimate probabilities

# Maximum Likelihood Estimation

Choose parameter $\theta$ that makes observed data most likely

$$\hat{\theta}_{MLE} = \arg\max_{\theta} P(\text{Data}|\theta)$$

**Example: Coin Bias**

100 flips: 60 heads, 40 tails. What's bias p?

$$P(\text{Data}|p) = \binom{100}{60} p^{60}(1-p)^{40}$$

$$\hat{p}_{MLE} = \frac{60}{100} = 0.6$$

**Result:** MLE estimate is the sample frequency




# Bayesian Parameter Estimation

Don't estimate a single "best" parameter. Compute the full **distribution** over parameters:

$$P(\theta|\text{Data}) = \frac{P(\text{Data}|\theta) \cdot P(\theta)}{P(\text{Data})}$$

01

### Prior P(θ)

Belief before seeing data

02

### Likelihood P(Data|θ)

How well θ explains data

03

### Posterior P(θ|Data)

Updated belief after data

**Advantages over MLE:** Quantifies uncertainty, incorporates prior knowledge, better with small datasets

# Key Takeaways

1. Modern AI is fundamentally probabilistic
2. Bayes' Rule is the most important formula

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

3. Base rates matter—avoid the base rate fallacy
4. Rational agents maximize expected utility under uncertainty
5. Probability represents degrees of belief, not just frequencies
6. Learning = belief updating with evidence